Pre-Read: Variables & Data Types + Arithmetic Operators

Nishut Suman 10 Nov, 2025 At 5:30 PM Pre-Reads Foundation Recommended Resource

Associated Lectures (1)

Description Discussions

🐍 PreRead: Variables, Data Types & Arithmetic Operators

🌱 1. What Are Variables?

Think of your computer's memory as a **big set of labeled containers (boxes)**. Each container stores a piece of information you want to use later. That label is called a **variable**.

🧠 Analogy

Imagine your **school bag** - each pocket has a label:

```
📘 books,
🍎 lunch_box,
🔑 key_chain
```

You know exactly what's inside each pocket. Variables do the same for your computer's memory!

💡 Example:

```
name = "Alice"
age = 21
height = 5.4
```

Variable	Value	Type
name	"Alice"	string
age	21	integer

Variable	Value	Type
height	5.4	float

🧾 2. Rules for Naming Variables

✅ Valid Names

```
user_name = "Bob"
age1 = 18
pi_value = 3.14
```

❌ Invalid Names

```
1user = "Tom"      # ❌ Can't start with a number
user-name = "Tom" # ❌ No hyphens
class = 10        # ❌ 'class' is a reserved keyword
```

Quick Tips:

- Use lowercase letters & underscores **(snake_case)**
- Be descriptive **(score, not x)**
- Python is **case-sensitive** `Age ≠ age ≠ AGe ≠ aGe`

🧩 3. Data Types in Python

Python automatically decides the **type** of data you store in a variable.

Data Type	Example	Description
int	a = 10	Whole numbers
float	b = 3.14	Numbers with decimals
str	name = "Alice"	Text or characters
bool	is_active = True	True or False values

🧪 Try It Out:

```
a = 10
b = 3.14
c = "Hello"
```

```
d = True

print(type(a))
print(type(b))
print(type(c))
print(type(d))
```

🖨️ **Output:**

```
<class 'int'>
<class 'float'>
<class 'str'>
<class 'bool'>
```

🔄 4. Type Casting (Changing Data Type)

You can **convert** one type into another - this is called **type casting**.

```
a = 10
print(float(a))   # 10.0
print(str(a))     # '10'
print(int(3.99))  # 3
```

Why do this? When taking input or doing calculations, you often need to ensure the correct type (e.g., converting text input to numbers).

🧬 5. Dynamic Typing

Python is **dynamically typed**, meaning you don't need to tell it what type of variable you're creating Python figures it out.

```
x = 10
print(type(x))   # int

x = "Hello"
print(type(x))   # str
```

🌀 The same variable can hold any data type at different times!

🗣️ 6. Input and Output

🖨️ Output using `print()`

```
name = "Alice"
print("Hello,", name)
```

Output:

```
Hello, Alice
```

🎤 Input using `input()`

```
name = input("Enter your name: ")
print("Hi,", name)
```

> ⚠️ Note: input() always returns text (a string). To use numbers, convert them:

```
age = int(input("Enter your age: "))
print("Next year you'll be", age + 1)
```

➕ 7. Arithmetic Operators

Operator	Description	Example	Output
+	Addition	5 + 3	8
-	Subtraction	5 - 3	2
*	Multiplication	5 * 3	15
/	Division (float)	5 / 2	2.5
%	Modulus (remainder)	5 % 2	1
//	Floor Division	5 // 2	2
**	Exponentiation (power)	2 ** 3	8

🧮 Example:

```
a = 10
b = 3
print(a + b)
print(a - b)
```

```
print(a * b)
print(a / b)
print(a % b)
print(a // b)
print(a ** b)
```

🧠 8. Real-Life Analogy

Situation	In Real Life	In Python
Counting apples	5 apples + 3 apples = 8	5 + 3
Sharing candies	10 candies / 4 friends	10 / 4
Full boxes only	10 // 4 = 2 full boxes	10 // 4
Remainder candies	10 % 4 = 2 left	10 % 4
Power of 2	$2^3 = 8$	2 ** 3

⚙️ 9. Mini Practice Challenge

🧩 Try this:

```
# Take two numbers as input and perform all operations
num1 = float(input("Enter first number: "))
num2 = float(input("Enter second number: "))

print("Addition:", num1 + num2)
print("Subtraction:", num1 - num2)
print("Multiplication:", num1 * num2)
print("Division:", num1 / num2)
print("Remainder:", num1 % num2)
print("Floor Division:", num1 // num2)
print("Power:", num1 ** num2)
```

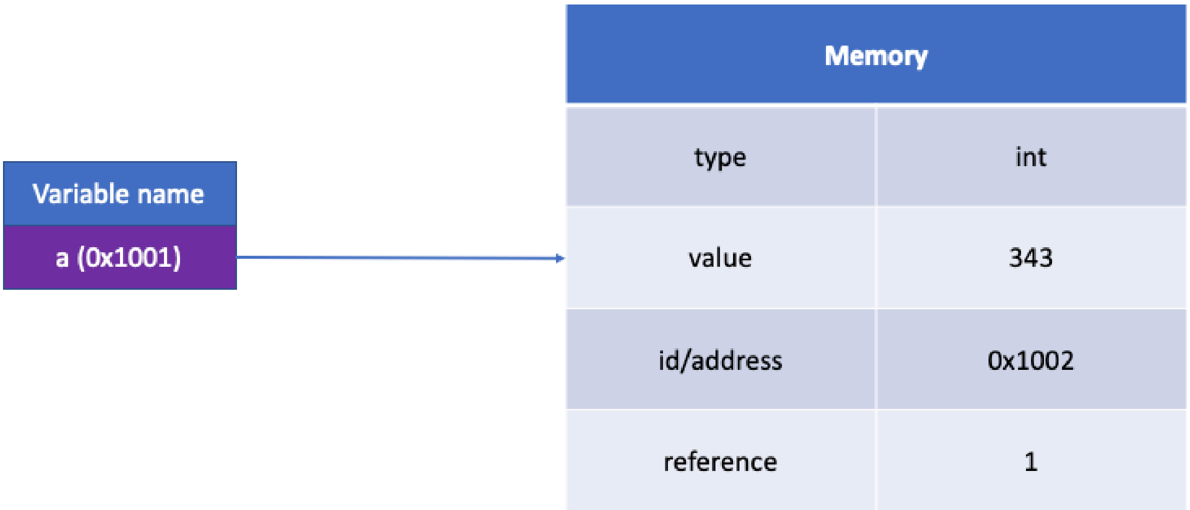
💡 **Bonus:** Print the type of each variable using `type()`

🚫 10. Common Beginner Mistakes (and Fixes!)

❌ Mistake	🤔 Why It Happens	✅ Correct Way
num = input("Enter: ") print(num + 5)	input() gives string, not number	num = int(input("Enter: "))

❌ Mistake	🤔 Why It Happens	✅ Correct Way
2name = "Bob"	Variable can't start with a number	name2 = "Bob"
print(2 ^ 3)	^ is **XOR**, not power	print(2 ** 3)
print(5 / 2) expecting integer	/ gives float	Use 5 // 2
Name = "A" print(name)	Python is case-sensitive	Match exact name (Name vs name)

🎨 Visual Concept Recap




```
Variable "a (0x1001)" is a name for data stored in memory.
Memory table shows: Type = "int" (whole number), Value = 343, Address = "0x1002", Referen
Arrow links "a" to its memory location.
This is how computers store and track data using names and addresses.
```

🪄 11. Quick Recap

```
✅ Variables → store values
✅ Data types → define what’s stored
✅ Type casting → change one type to another
✅ Dynamic typing → Python figures out types for you
✅ print() & input() → communication tools
✅ Arithmetic operators → perform math logic
```

🏁 Final Thought

> “Variables are like memory notes you leave for your computer. You tell it what to remember - and Python never forgets (until you overwrite it)!”